

Binomial Coefficients

Section 2.2

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



$$2 + 3 + 2 = 7$$

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Today's lecture

1. Basic Identities

2. The Binomial Theorem

3. Double Counting

4. Generalized Coefficients

One number, several interpretations

$$\binom{n}{k}$$

counts k -element subsets of an n -element set, binary words of length n with k ones, and lattice paths with k up steps and $n - k$ right steps.

Each interpretation suggests different identities.

Conventions

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (0 \leq k \leq n).$$

For nonnegative integer n , use

$$\binom{n}{k} = 0 \quad \text{if } k < 0 \text{ or } k > n.$$

Also $\binom{n}{0} = \binom{n}{n} = 1$, including $\binom{0}{0} = 1$.

Identity

$$\binom{n}{k} = \binom{n}{n-k}.$$

Proof

Taking complements is a bijection from k -subsets to $(n - k)$ -subsets of a fixed n -element set.

Pascal's identity

Identity

For $n \geq 1$,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Proof

Fix one element a . A k -subset either omits a , or contains a and $k-1$ of the other elements. These two cases are disjoint.

Pascal's triangle

n								
0	1							
1	1	1						
2	1	2	1					
3	1	3	3	1				
4	1	4	6	4	1			
5	1	5	10	10	5	1		
6	1	6	15	20	15	6	1	

Each interior entry is the sum of the two entries immediately above it.

The binomial theorem

Binomial theorem

For every nonnegative integer n ,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

The coefficients of a familiar algebraic expansion count choices.

A combinatorial proof

Proof

Expand the product of n factors $(x + y)$. To obtain $x^{n-k}y^k$, choose the k factors that contribute y ; all others contribute x .

There are $\binom{n}{k}$ such choices, which is exactly the coefficient of $x^{n-k}y^k$.

An example

$$(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5.$$

Question

What is the coefficient of x^3y^4 in $(2x - y)^7$?

It is

$$\binom{7}{4} 2^3 (-1)^4 = 280.$$

Identity

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

Proof

Set $x = y = 1$ in the binomial theorem. Alternatively, count all subsets by their sizes.

Alternating row sums

Identity

For $n \geq 1$,

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

Proof

Set $x = 1, y = -1$. A combinatorial proof pairs each subset with the subset obtained by toggling one fixed element. Each pair contains one even-sized and one odd-sized subset.

Even and odd subsets

$$\sum_{\substack{k=0 \\ k \text{ even}}}^n \binom{n}{k} = \sum_{\substack{k=0 \\ k \text{ odd}}}^n \binom{n}{k} = 2^{n-1} \quad (n \geq 1).$$

Add and subtract the ordinary row sum and alternating row sum.

A distinguished element

Identity

$$k \binom{n}{k} = n \binom{n-1}{k-1}.$$

Proof

Count pairs (S, a) where S is a k -subset and $a \in S$. Choose S then a , or choose a then the other $k - 1$ elements of S .

A weighted row sum

Identity

For $n \geq 1$,

$$\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}.$$

Proof

Count a subset together with one distinguished member. Choose that member in n ways, then choose any subset of the other $n - 1$ elements.

Two distinguished elements

Identity

For $n \geq 2$,

$$\sum_{k=0}^n k(k-1) \binom{n}{k} = n(n-1)2^{n-2}.$$

Proof

Count a subset with an ordered pair of distinct distinguished members. Choose those members first, then any subset of the remaining $n - 2$ elements.

The sum involving k^2

Since $k^2 = k(k-1) + k$, for $n \geq 2$,

$$\sum_{k=0}^n k^2 \binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}.$$

The identity for a more complicated weight follows from simpler counting interpretations.

Vandermonde's identity

Identity

For nonnegative integers r, s, m ,

$$\binom{r+s}{m} = \sum_{k=0}^m \binom{r}{k} \binom{s}{m-k}.$$

Proof

Split a set into parts of sizes r and s . Count its m -subsets according to the number k chosen from the first part.

Vandermonde by coefficients

Proof

The identity

$$(1+x)^{r+s} = (1+x)^r(1+x)^s$$

gives two expressions for the coefficient of x^m . On the right, choose x^k from the first factor and x^{m-k} from the second, and sum over k .

This is our first example of a convolution of two sequences.

A sum of squares

Identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

Proof

Use Vandermonde with $r = s = m = n$ and symmetry:

$$\binom{n}{n-k} = \binom{n}{k}.$$

The hockey-stick identity

Identity

For $0 \leq r \leq n$,

$$\sum_{j=r}^n \binom{j}{r} = \binom{n+1}{r+1}.$$

Proof

Count $(r+1)$ -subsets of $\{1, \dots, n+1\}$ according to their largest element $j+1$. The other r elements can be chosen in $\binom{j}{r}$ ways.

Nested subsets

Identity

For $0 \leq r \leq k \leq n$,

$$\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}.$$

Proof

Count pairs $R \subseteq K \subseteq [n]$ with $|R| = r$ and $|K| = k$. Choose the larger set first, or choose the smaller set first.

Generalized binomial coefficients

Definition

For a number x and a nonnegative integer k , define

$$\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}, \quad \binom{x}{0} = 1.$$

This is a polynomial in x . It need not count subsets when x is not a nonnegative integer.

Negative upper arguments

Identity

For positive integer m and $k \geq 0$,

$$\binom{-m}{k} = (-1)^k \binom{m+k-1}{k}.$$

Proof

Factor -1 from each of the k numerator factors:

$$\frac{(-m)(-m-1)\cdots(-m-k+1)}{k!} = (-1)^k \frac{m(m+1)\cdots(m+k-1)}{k!}.$$

Pascal's identity still holds

Identity

For $k \geq 1$ and any x ,

$$\binom{x}{k} = \binom{x-1}{k} + \binom{x-1}{k-1}.$$

Proof

Both sides are polynomials in x . They agree for all sufficiently large integers x by Pascal's identity, and therefore agree identically.

A useful infinite expansion

$$(1 - z)^{-m} = \sum_{k \geq 0} \binom{m + k - 1}{k} z^k \quad (m \geq 1).$$

As a numerical series this holds for $|z| < 1$. As a formal power series it is an identity of coefficients.

We will recover these coefficients when counting selections with repetition.

Question

Prove

$$\sum_{k=r}^n \binom{n}{k} \binom{k}{r} = \binom{n}{r} 2^{n-r}.$$

Count a subset $S \subseteq [n]$ together with a distinguished r -subset $R \subseteq S$.

Proof

On the left, first choose $|S| = k$, then S , then R .

On the right, first choose R in $\binom{n}{r}$ ways. Each of the remaining $n - r$ elements is independently either included in S or excluded.

Both procedures count every pair (R, S) exactly once.

- Binomial coefficients count subsets and appear as expansion coefficients.
- Pascal, Vandermonde, and hockey-stick identities have direct counting proofs.
- Distinguished elements produce weighted sums.
- Generalized coefficients extend the algebra beyond nonnegative upper arguments.

Multinomial Coefficients